

8	(a)	Evidence for the existence of the nucleus came from the observations of how α -particles were scattered by thin metal foil in experiments.		
		(i)	State the observation that provided evidence for	
			1. the small size of the nucleus as compared to that of the atom, 	[1]
		1.	Solution: Most α -particles were deviated through small angles.	1
			2. the nucleus being positively charged and containing the majority of the mass of the atom. 	[2]
		2.	Solution: Small fraction of α -particles were deviated through large angles, greater than 90° .	1 1
		(ii)	For the experiment, suggest why	
			1. α-particles are used, 	[2]
		1.	Solution: They are positively charged, and hence will undergo electrostatic interaction with the target nuclei. They are produced naturally by radioactive nuclides that are α -particle emitters, and so it was easy to obtain a steady supply of α -particles. α -particles emitted from the same type of radioactive nuclide have almost the same kinetic energy. Thus it was easy to keep this control variable constant in the experiment. (They have small masses, and hence will be scattered easily.)	1 1

			2. a thin metal foil is used. 		[1]									
		2.	Solution: A thin metal foil is used so as to ensure that one α-particle collides with a single gold nucleus. Or to avoid multiple collisions of one α-particle with many gold nuclei.		1									
	(b)	The binding energy of various nuclei is commonly being determined in the study of nuclear sources.												
		(i)	Use the data below to determine, to five significant figures, the binding energy of a nucleus of $^{16}_8\text{O}$. Data: <table><tr><td>mass of proton</td><td>=</td><td>1.007276 u</td></tr><tr><td>mass of neutron</td><td>=</td><td>1.008665 u</td></tr><tr><td>mass of oxygen nucleus</td><td>=</td><td>15.990527 u</td></tr></table> binding energy = MeV	mass of proton	=	1.007276 u	mass of neutron	=	1.008665 u	mass of oxygen nucleus	=	15.990527 u		[3]
mass of proton	=	1.007276 u												
mass of neutron	=	1.008665 u												
mass of oxygen nucleus	=	15.990527 u												
		(i)	Solution: Mass defect, $\Delta m = 8(1.007276) + 8(1.008665) - 15.990527 = 0.137001 \text{ u}$ Binding energy, $\Delta mc^2 = (0.137001 \times 1.66 \times 10^{-27})(3.00 \times 10^8)^2$ $= 2.04679 \times 10^{-11} \text{ J}$ $= \frac{2.04679 \times 10^{-11}}{1.60 \times 10^{-19}} \text{ MeV}$ $= 127.92 \text{ MeV (5 s.f.)}$		1 1 1									
		(ii)	The binding energy of $^{17}_8\text{O}$ is found to be 126.43 MeV. State and explain which of these two isotopes of oxygen, $^{16}_8\text{O}$ or $^{17}_8\text{O}$ would be more stable.											

				[3]
		(ii)	Solution: Binding energy per nucleon for $^{17}_8\text{O} = \frac{126.43}{17} = 7.44 \text{ MeV}.$ Binding energy per nucleon for $^{16}_8\text{O} = \frac{127.92}{16} = 8.00 \text{ MeV}.$ Since $^{16}_8\text{O}$ has a higher binding energy per nucleon, it would be more stable.	1 1 1
	(c)	Uranium-234 is an isotope of uranium that decays by α -emission to form thorium-230. The average half-life of uranium-234 is about 2.5×10^5 years.		
		(i)	Explain why the half-life is an average value. 	[2]
		(i)	Solution: This is because radioactive decay is a spontaneous and random process. The exact value of the number of undecayed nuclei present at that time fluctuates/cannot be determined with certainty. (The sample size is large.)	1 1
		(ii)	In a sample of radioactive material, the ratio of uranium atoms to thorium atoms is 1:6.	
			1. Estimate the age of the sample, assuming that originally there is no thorium present. age = years	[3]

